PRACTICAL NUMBER: 3

AIM: Program based on linked list:

Maintain a record of coffee shop database implement following:

i. Display the total sale of a day

ii. Maximum and minimum bill of a day

***PROGRAM*:**

```
//sheikh wasimuddin cse(iot) 60  23071044 //
#include <stdio.h>
#include <stdlib.h>
typedef struct CoffeeShop {
   int billAmount;
   struct CoffeeShop* next;
} CoffeeShop;
void insertBill(CoffeeShop** head, int amount);
int getTotalSales(CoffeeShop* head);
int getSalesCount(CoffeeShop* head);
int getMaxBill(CoffeeShop* head);
int getMinBill(CoffeeShop* head);
void freeList(CoffeeShop* head);
int main() {
   CoffeeShop* head = NULL;
   int choice, amount;
   do {
```

```
    printf("\n1. Add bill\n2. Display total sales\n3. Display number of sales\n4. Display maximum bill\n5. Display minimum bill\n6. Exit\n");
    printf("Enter your choice: ");
    scanf("%d", &choice);
    switch(choice) {
        case 1:
            printf("Enter bill amount: ");
            scanf("%d", &amount);
            insertBill(&head, amount);
            break;
        case 2:
            printf("Total sales of the day: %d\n", getTotalSales(head));
            break;
        case 3:
            printf("Number of sales: %d\n", getSalesCount(head));
            break;
        case 4:
            printf("Maximum bill of the day: %d\n", getMaxBill(head));
            break;
        case 5:
            printf("Minimum bill of the day: %d\n", getMinBill(head));
            break;
        case 6:
            freeList(head);
```

```
            printf("Exiting...\n");
            break;
        default:
            printf("Invalid choice, please try again.\n");
    }
  } while (choice != 6);
  return 0;
}
void insertBill(CoffeeShop** head, int amount) {
  CoffeeShop* newNode = (CoffeeShop*)malloc(sizeof(CoffeeShop));
  newNode->billAmount = amount;
  newNode->next = NULL;
  if (*head == NULL) {
    *head = newNode;
  } else {
    CoffeeShop* temp = *head;
    while (temp->next != NULL) {
      temp = temp->next;
    }
    temp->next = newNode;
  }
}
int getTotalSales(CoffeeShop* head) {
  int total = 0;
```

```
    CoffeeShop* temp = head;
    while (temp != NULL) {
        total += temp->billAmount;
        temp = temp->next;
    }
    return total;
}
int getSalesCount(CoffeeShop* head) {
    int count = 0;
    CoffeeShop* temp = head;
    while (temp != NULL) {
        count++;
        temp = temp->next;
    }
    return count;
}
int getMaxBill(CoffeeShop* head) {
    if (head == NULL) {
        return 0;
    }
    int max = head->billAmount;
    CoffeeShop* temp = head->next;
    while (temp != NULL) {
        if (temp->billAmount > max) {
```

```
            max = temp->billAmount;
        }
        temp = temp->next;
    }
    return max;
}
int getMinBill(CoffeeShop* head) {
    if (head == NULL) {
        return 0;
    }
    int min = head->billAmount;
    CoffeeShop* temp = head->next;
    while (temp != NULL) {
        if (temp->billAmount < min) {
            min = temp->billAmount;
        }
        temp = temp->next;
    }
    return min;
}
void freeList(CoffeeShop* head) {
    CoffeeShop* temp;
    while (head != NULL) {
        temp = head;
```

```
        head = head->next;
        free(temp);
    }
}
```

OUTPUT:

main.c

```c
//sheikh wasimuddin cse(iot) 60  23071044 //
#include <stdio.h>
#include <stdlib.h>

typedef struct CoffeeShop {
    int billAmount;
    struct CoffeeShop* next;
} CoffeeShop;

void insertBill(CoffeeShop** head, int amount);
int getTotalSales(CoffeeShop* head);
int getSalesCount(CoffeeShop* head);
int getMaxBill(CoffeeShop* head);
int getMinBill(CoffeeShop* head);
void freeList(CoffeeShop* head);

int main() {
    CoffeeShop* head = NULL;
    int choice, amount;

    do {
        printf("\n1. Add bill\n2. Display total sales\n3. Display number
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("Enter bill amount: ");
                scanf("%d", &amount);
                insertBill(&head, amount);
                break;
            case 2:
                printf("Total sales of the day: %d\n", getTotalSales(head
                break;
            case 3:
```

```
1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 1
Enter bill amount: 5000

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 1
Enter bill amount: 6000

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 2
Total sales of the day: 11000

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 3
Number of sales: 2

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
```

main.c

```c
//sheikh wasimuddin cse(iot) 60  23071044 //
#include <stdio.h>
#include <stdlib.h>

typedef struct CoffeeShop {
    int billAmount;
    struct CoffeeShop* next;
} CoffeeShop;

void insertBill(CoffeeShop** head, int amount);
int getTotalSales(CoffeeShop* head);
int getSalesCount(CoffeeShop* head);
int getMaxBill(CoffeeShop* head);
int getMinBill(CoffeeShop* head);
void freeList(CoffeeShop* head);

int main() {
    CoffeeShop* head = NULL;
    int choice, amount;

    do {
        printf("\n1. Add bill\n2. Display total sales\n3. Display number
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("Enter bill amount: ");
                scanf("%d", &amount);
                insertBill(&head, amount);
                break;
            case 2:
                printf("Total sales of the day: %d\n", getTotalSales(head
                break;
            case 3:
```

```
Total sales of the day: 11000

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 3
Number of sales: 2

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 4
Maximum bill of the day: 6000

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 5
Minimum bill of the day: 5000

1. Add bill
2. Display total sales
3. Display number of sales
4. Display maximum bill
5. Display minimum bill
6. Exit
Enter your choice: 6
Exiting...


...Program finished with exit code 0
Press ENTER to exit console.
```

CONCLUSION: This practical was performed successfully